MAXIMILIAN LIPSKI

Rendering Programmer | C++, Vulkan | Real-Time Rendering

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SUMMARY

Rendering professional with half a year of experience in machine learning research and real-time rendering, expert in C++ and Vulkan. Key achievements include developing a real-time rendering engine achieving around 400 FPS at 2560×1600 resolution and building a Vulkan RHI with future backend support capabilities.

EXPERIENCE

Student Research Assistant

TU Dresden - ScaDS.AI

11/2025 - 04/2026 · Dresden, DE

http://scads.ai

- Researched machine learning approaches to accelerate geometric rendering of geometric objects
- Developed and trained neural networks for geometric approximation problems using CUDA-enabled machine learning workflows
- Investigated neural representations of Hermite spline tubes as signed distance functions
- Trained a neural network for ray-intersection with Hermite spline tubes
- Designed and evaluated experiments to analyze the accuracy and performance of learned geometric approximations
- Achieved promising results for fixed primitives while identifying parameterization as a major challenge or generalization

PROJECTS

Rodan - Real-Time Rendering Engine

- Created a real-time rendering engine built on top of a custom Rendering-Hardware Interface (RHI) with a Vulkan backend
- Achieves around 400 FPS while rendering the Sponza scene at 2560×1600 on an RTX 5070
- Built a physically based rendering (PBR) pipeline with image-based lighting (IBL)
- Developed an HDR tone mapping pipeline
- Added glTF 2.0 asset loading with support for KHR_materials_transmission and KHR_materials_volume
- Integrated Dear ImGui for in-engine tooling and debugging
- Debugged and profiled rendering workloads using NVIDIA Nsight Graphics and RenderDoc

Velos - Vulkan RHI

- Designed and implemented a backend-independent RHI with a Vulkan backend
- Developed abstractions for pipelines, descriptor sets, textures, samplers, buffers, and command lists
- Built automatic SPIR-V shader reflection for descriptor set and pipeline layout generation
- Architected the system to support future graphics backends such as Direct3D 12

EDUCATION

Master of Science, Computer Science

Technical University Dresden

Dresden, DE

- Specialization: Computer Graphics
- Expected Graduation: 01/2027

Bachelor of Science, Computer Science

Heinrich-Heine University

10/2020 - 09/2024 · Düsseldorf, DE

- Grade: 1.9
- Thesis: Development of a Skew-Heap Data Structure for Persistent Memory

SKILLS

Programming

C/C++ · GLSL · CUDA

Graphics APIs

Vulkan · OpenGL

Tools

RenderDoc · NVIDIA Nsight Graphics · NVIDIA Nsight Systems · Git · CMake · Premake · Unreal Engine 5

Platforms

Linux · Windows

LANGUAGES

German
Native ●●●●●

Polish
Native ●●●●●

English
C1 ●●●●○